QuantForge

Multi-Mode Adaptive Strategy Report

2026-04-29

Trend Follow + High-Vol Breakout + Wheel (Options) + Cash

11 Symbols x 10Y backtest

Strategy Architecture

Strategy Architecture: Parallel Trend + Wheel System

```
Two independent engines run simultaneously:

ENGINE 1 - Trend/Breakout (fires when signal present):
  if regime == BEAR/CRISIS       -> CASH (no new entries)
  elif BULL + vol > 45%          -> HIGHVOL_BREAKOUT (4x ATR)
  else (BULL/CONSOL/NEUTRAL)     -> TREND_FOLLOW (3x ATR)

ENGINE 2 - Wheel (runs in parallel during consolidation):
  if regime == CONSOLIDATION/NEUTRAL -> WHEEL (CSP/CC)
  else                               -> wind down

Multiple trend positions can be open simultaneously.
Total exposure capped at 80% of capital.
```

Regime-Adaptive Position Sizing

```
BULL_TREND:     50% base, 10% risk cap
NEUTRAL:        30% base, 5% risk cap
CONSOLIDATION:  20% base, 3% risk cap
```

Exit Rules (all modes)

```
Trailing stop:  3x ATR (trend) / 4x ATR (breakout)
Regime exit:    bull->bear/crisis after 5 bars
MA breakdown:   2 consecutive closes below entry MA
Time limit:     250d max hold
```

Wheel Parameters

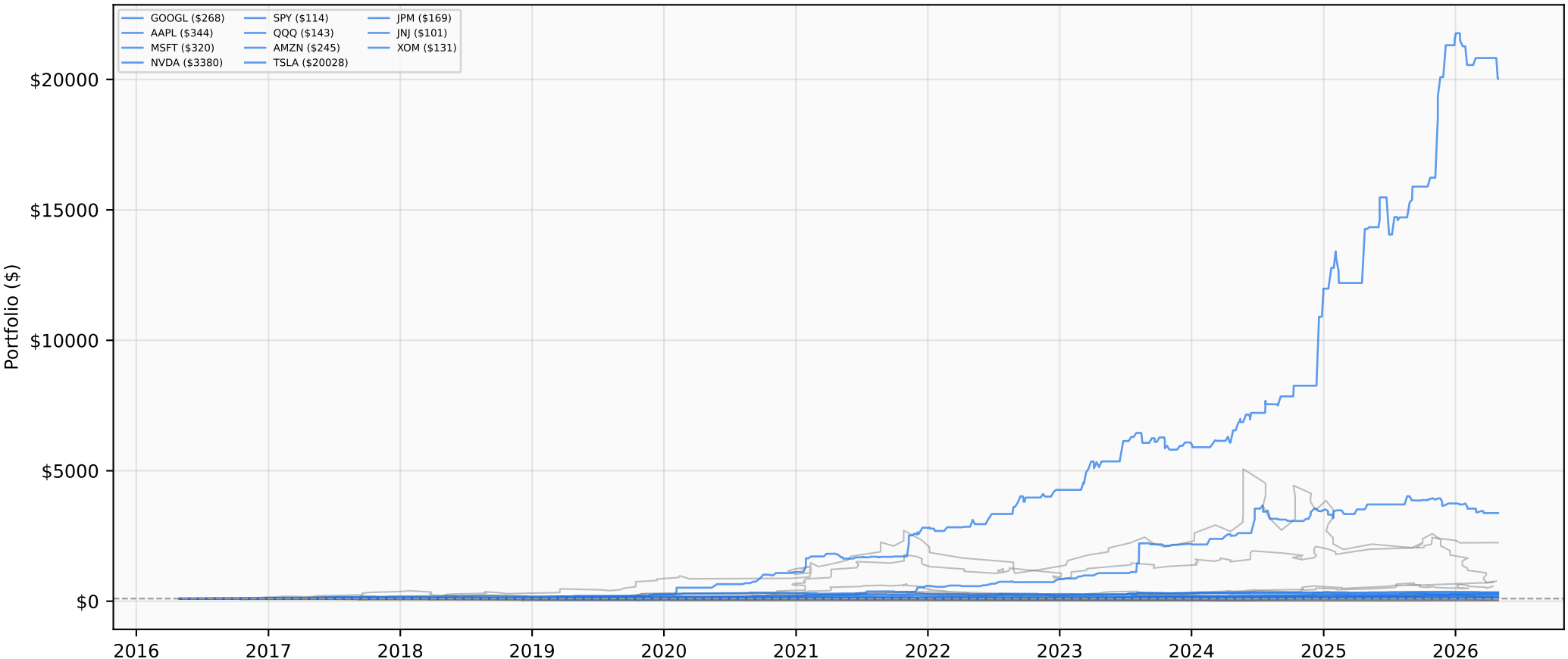
```
Cycle:          30 trading days (~45 calendar)
OTM:            10%
```

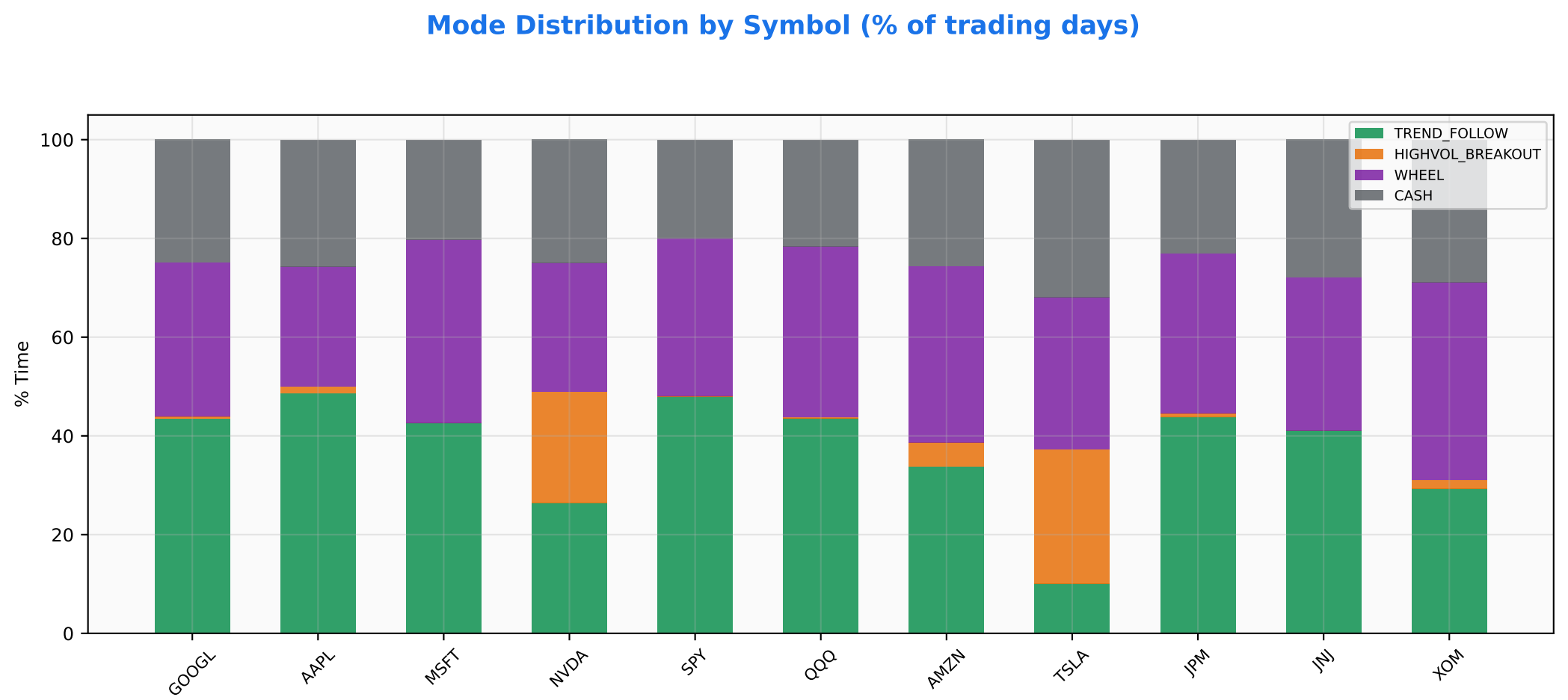
```
Premium disc:   85% of BSM
Commission:     $0.65/contract
```

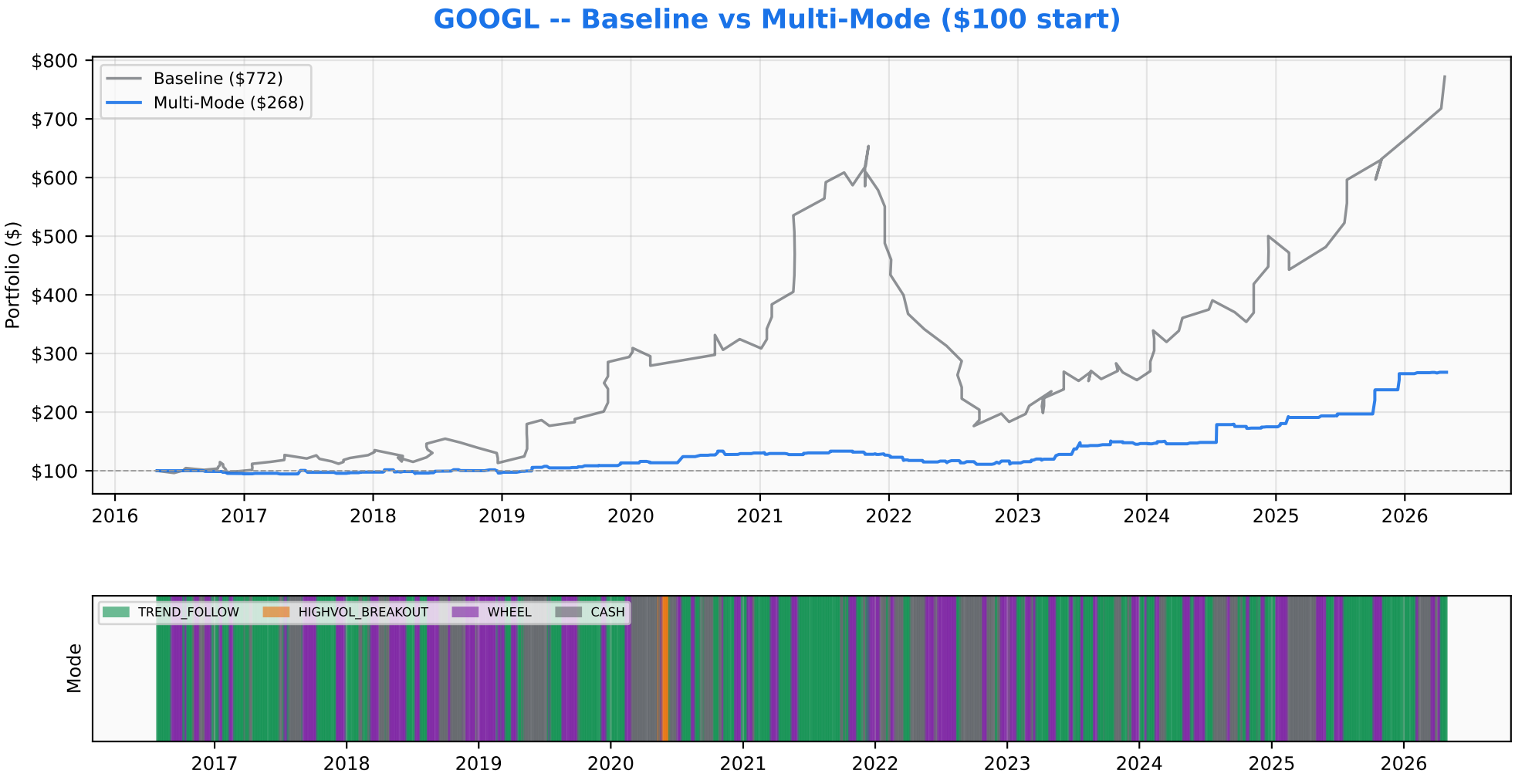
Multi-Mode vs Baseline Summary

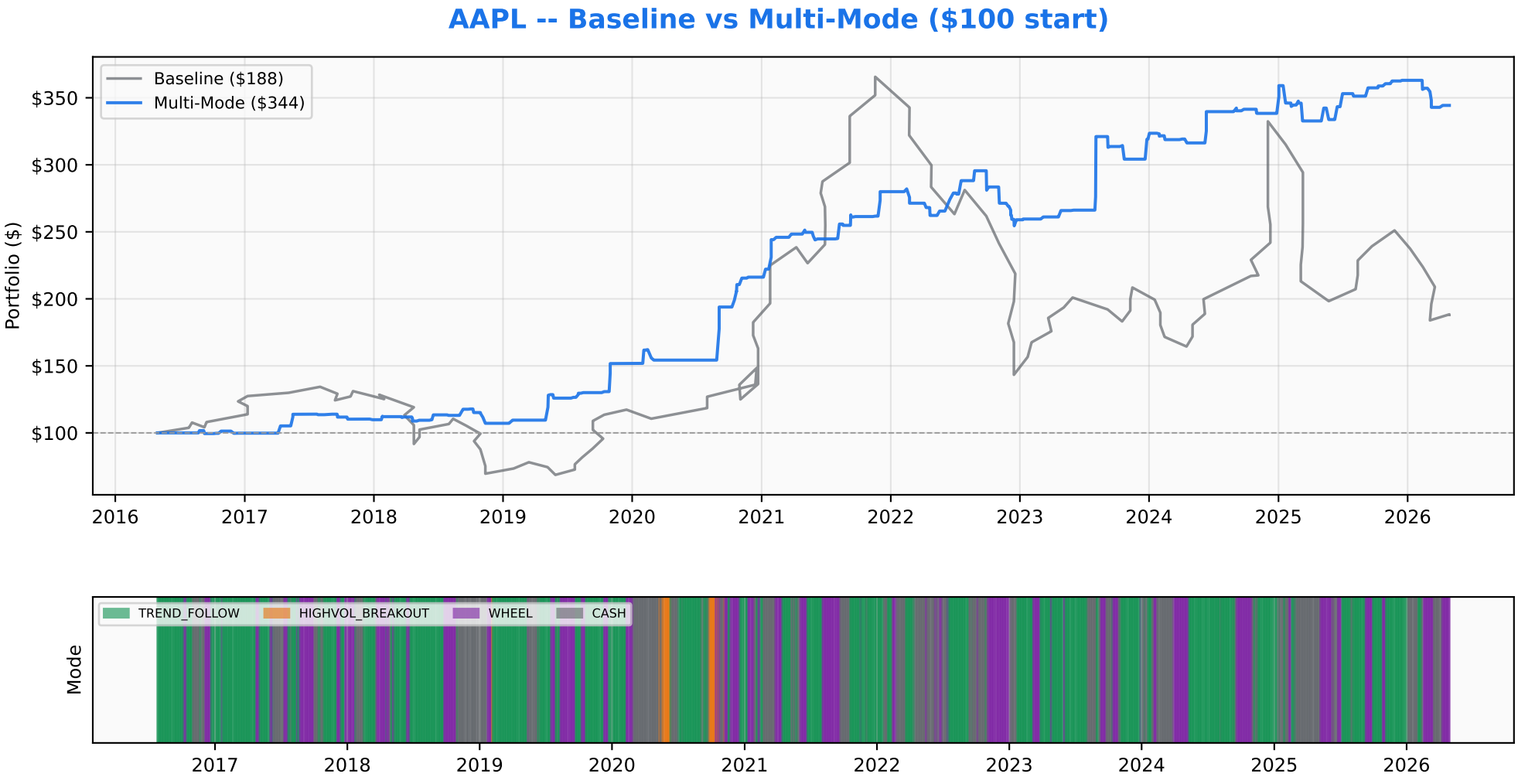
Symbol	Old$	New$	OldSharpe	NewSharpe	Winner	PrimaryMode
GOOGL	$772	$268	3.22	1.95	OLD	TREND_FOLLOW
AAPL	$188	$344	1.27	2.40	NEW	TREND_FOLLOW
MSFT	$2247	$320	4.64	1.59	OLD	TREND_FOLLOW
NVDA	$760	$3380	2.35	2.15	NEW	TREND_FOLLOW
SPY	$53	$114	-1.24	-0.32	NEW	TREND_FOLLOW
QQQ	$176	$143	1.34	0.22	OLD	TREND_FOLLOW
AMZN	$576	$245	2.56	1.12	OLD	WHEEL
TSLA	$158	$20028	1.02	2.35	NEW	CASH
JPM	$43	$169	-0.88	-0.40	NEW	TREND_FOLLOW
JNJ	$20	$101	-3.22	-0.78	NEW	TREND_FOLLOW
XOM	$21	$131	-1.83	-1.07	NEW	WHEEL

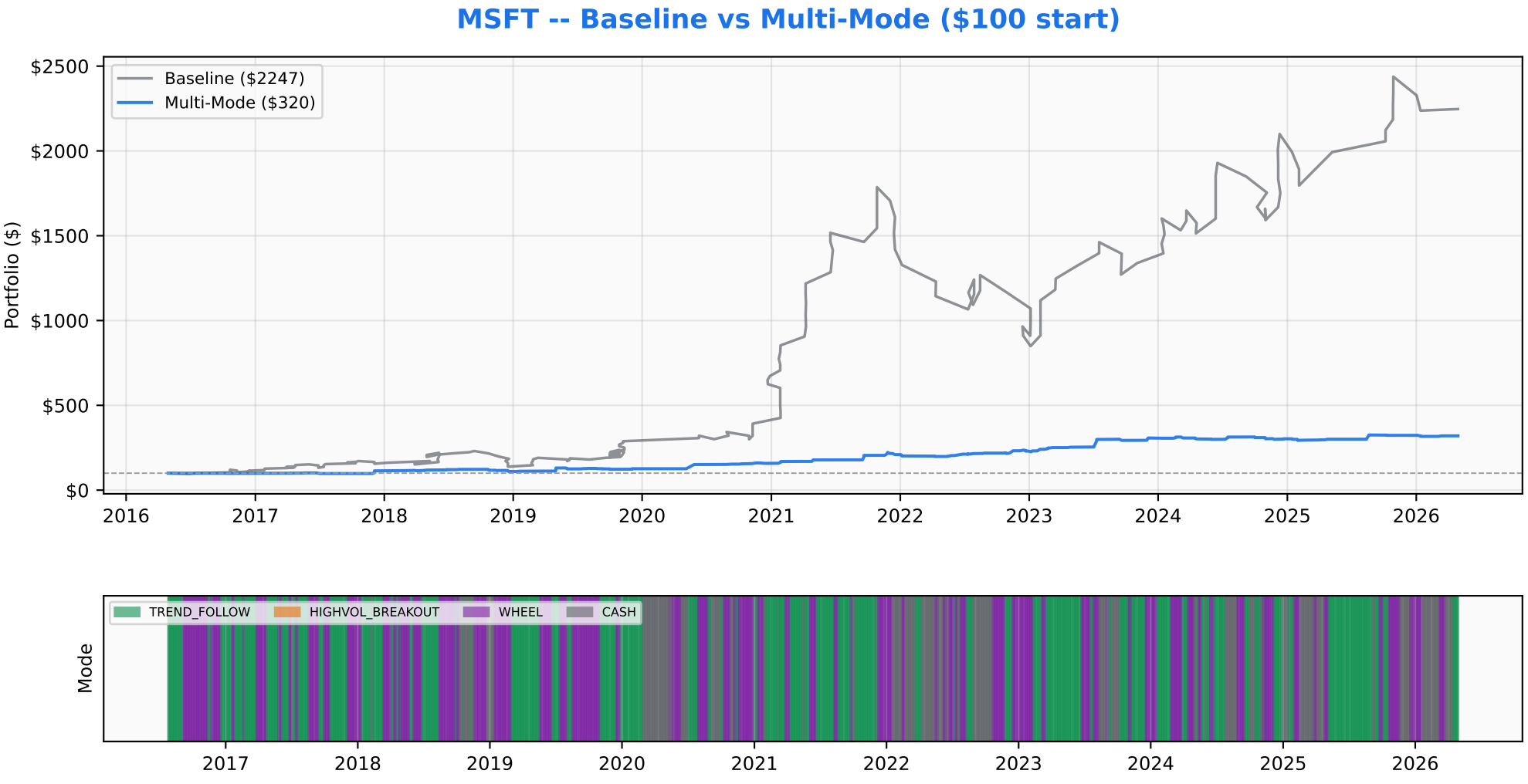
All Symbols: Baseline (gray) vs Multi-Mode (blue)

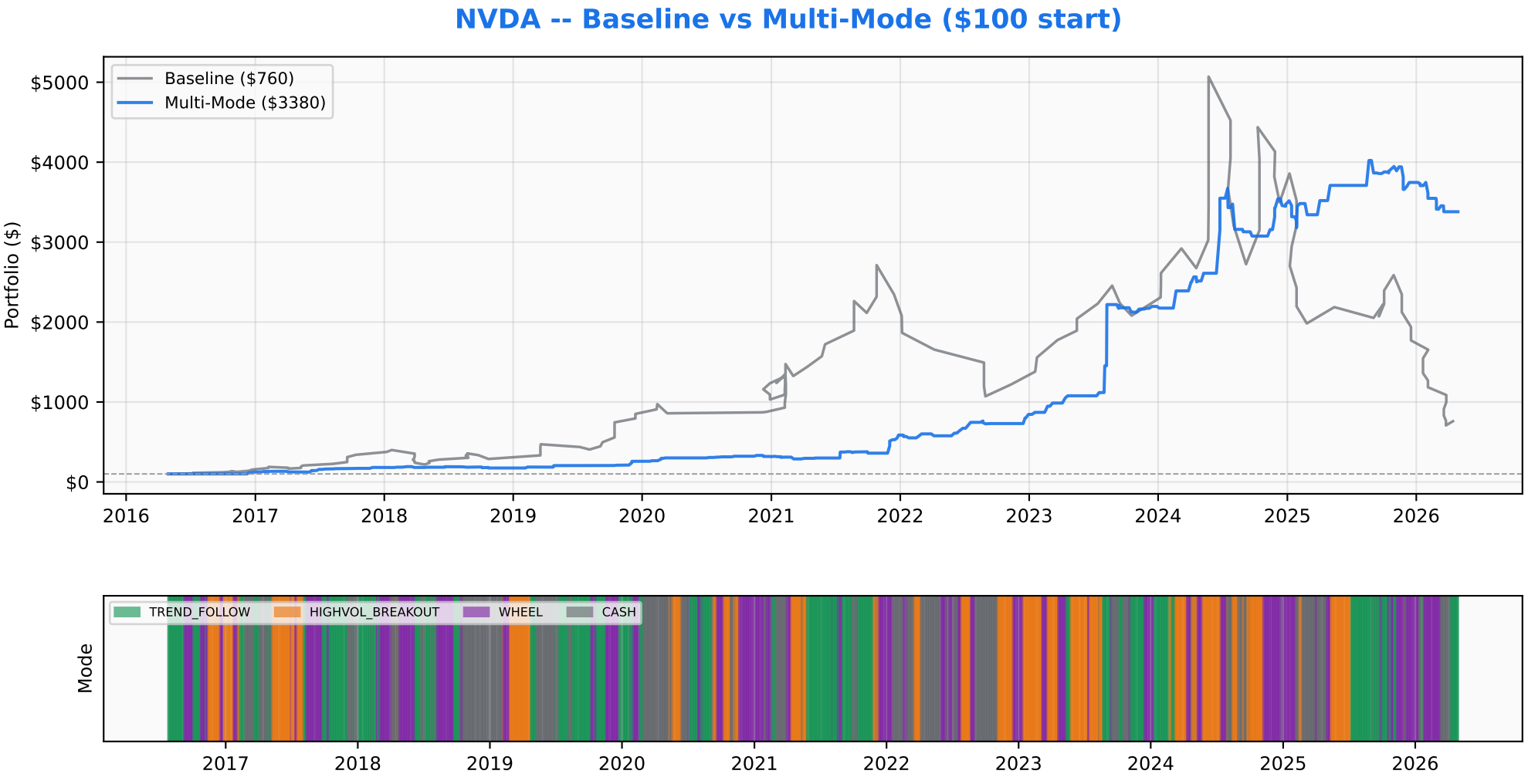



Mode Distribution by Symbol (% of trading days)
% Time
0
20
40
60
80
100
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
GOOGL
AAPL
MSFT
NVDA
SPY
QQQ
AMZN
TSLA
JPM
JNJ
XOM


GOOGL -- Baseline vs Multi-Mode ($100 start)
Baseline ($772)
Multi-Mode ($268)
Portfolio ($)
$800
$700
$600
$500
$400
$300
$200
$100
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
Mode
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026


AAPL -- Baseline vs Multi-Mode ($100 start)
Baseline ($188)
Multi-Mode ($344)
Portfolio ($)
$350
$300
$250
$200
$150
$100
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026


MSFT -- Baseline vs Multi-Mode ($100 start)
Baseline ($2247)
Multi-Mode ($320)
Portfolio ($)
$2500
$2000
$1500
$1000
$500
$0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH


NVDA -- Baseline vs Multi-Mode ($100 start)
Baseline ($760)
Multi-Mode ($3380)
Portfolio ($)
$5000
$4000
$3000
$2000
$1000
$0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
2017
2018
2019
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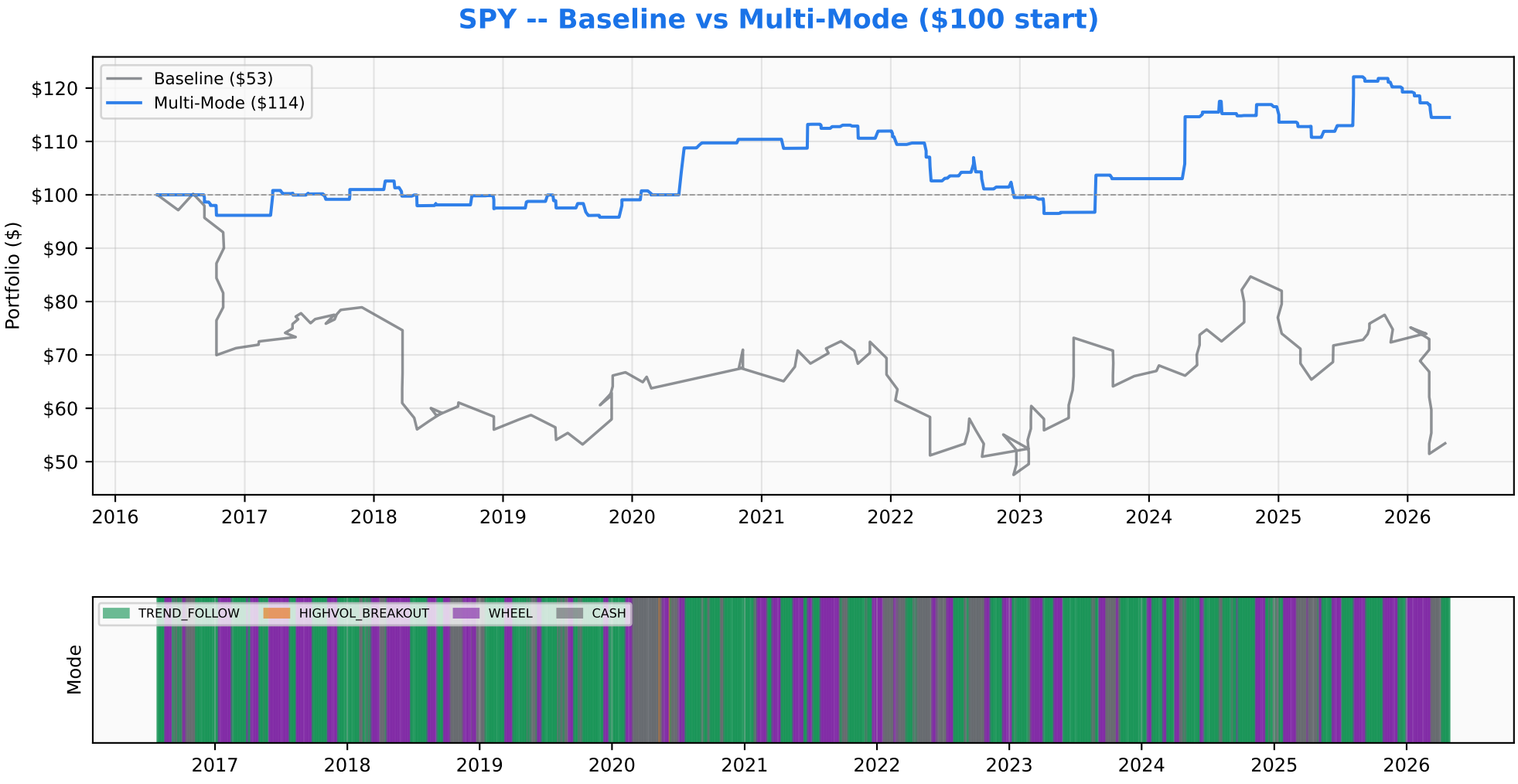
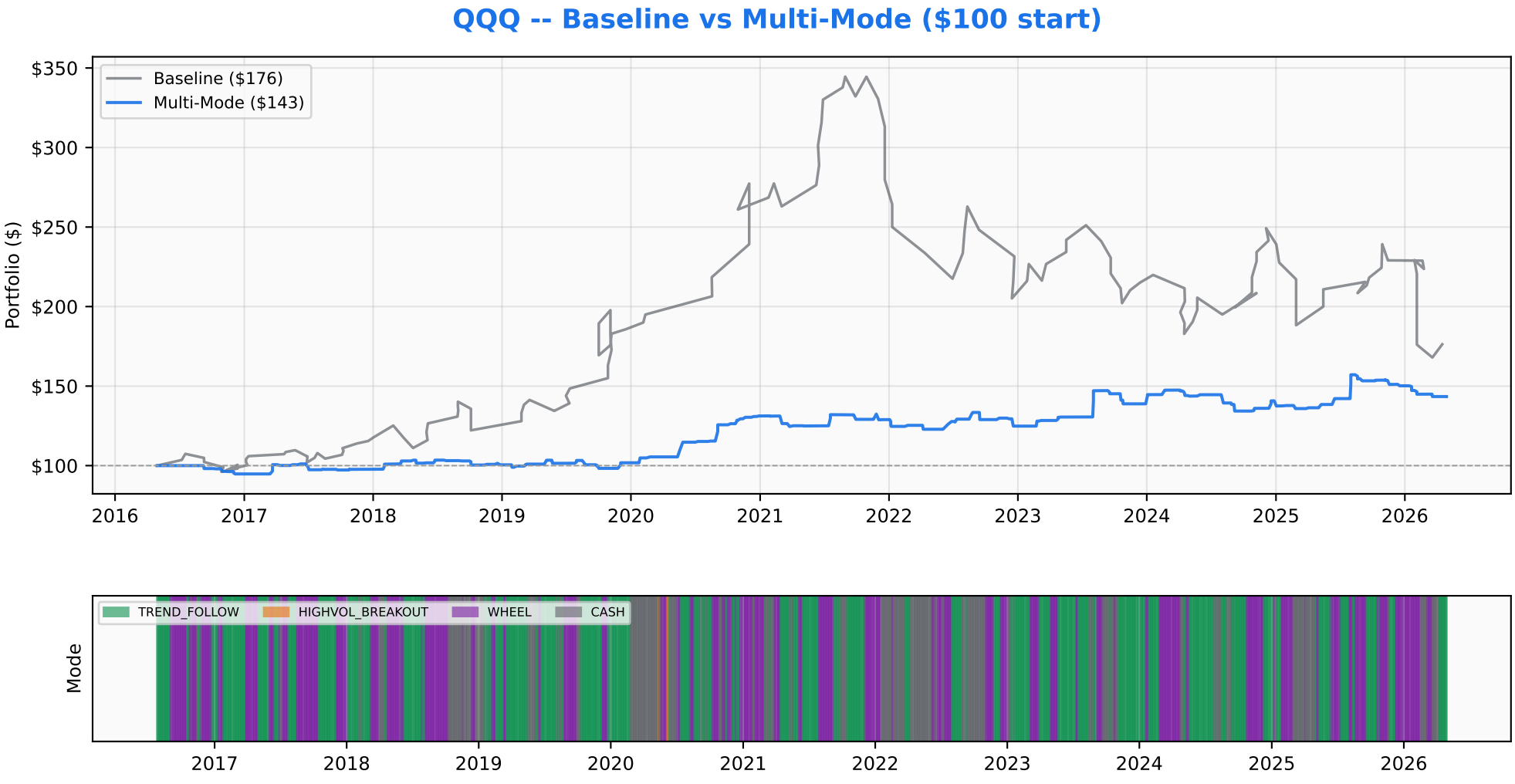
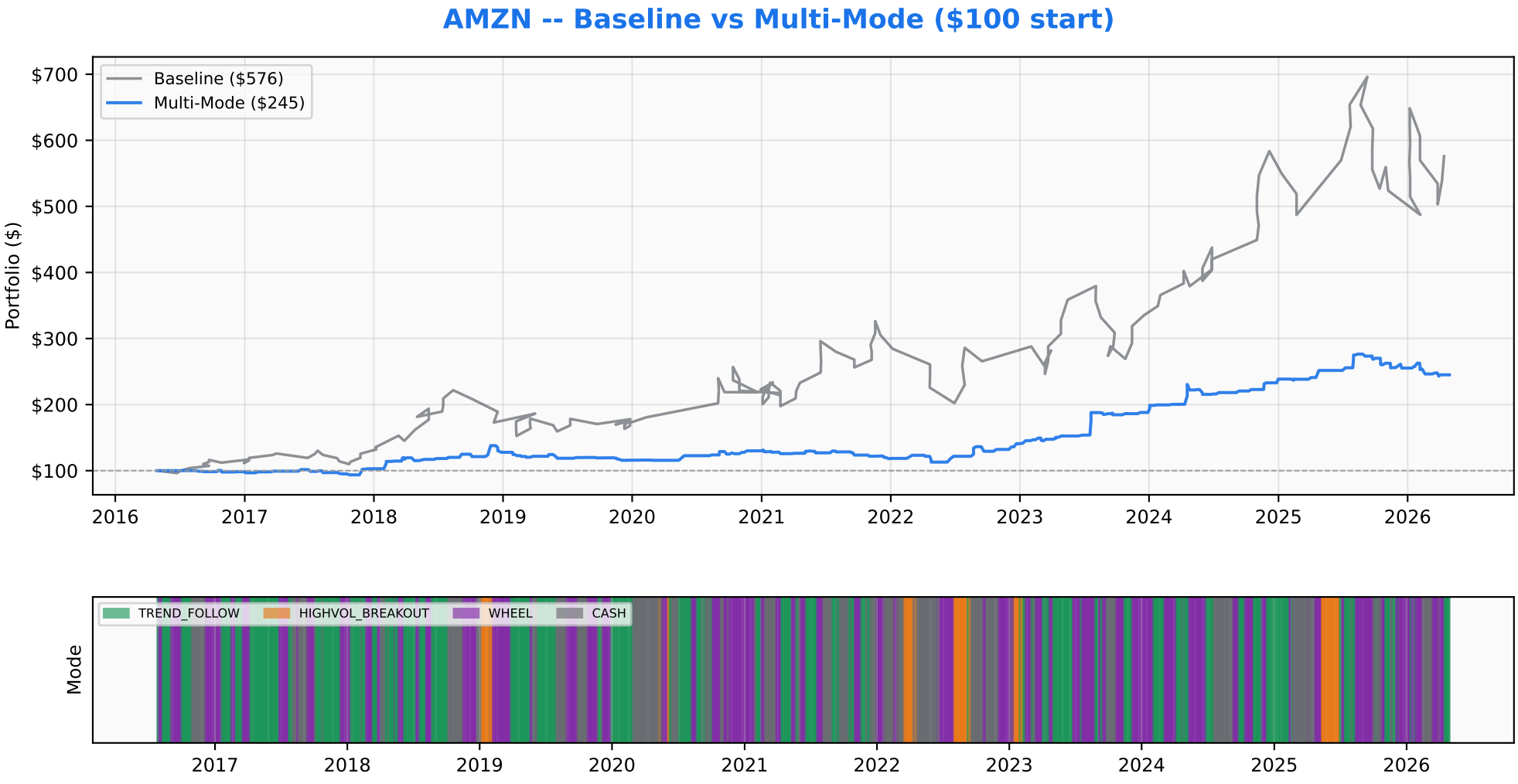
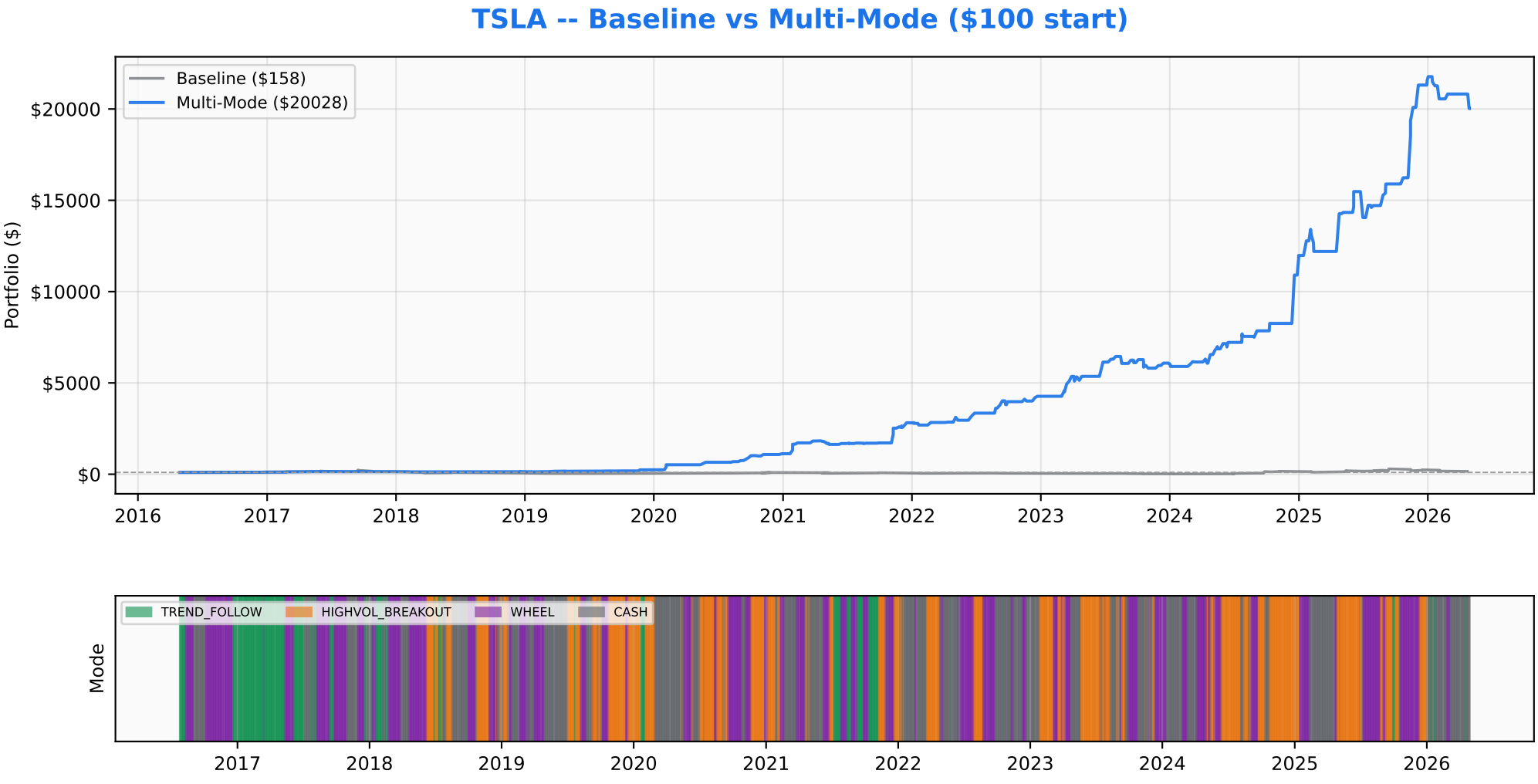
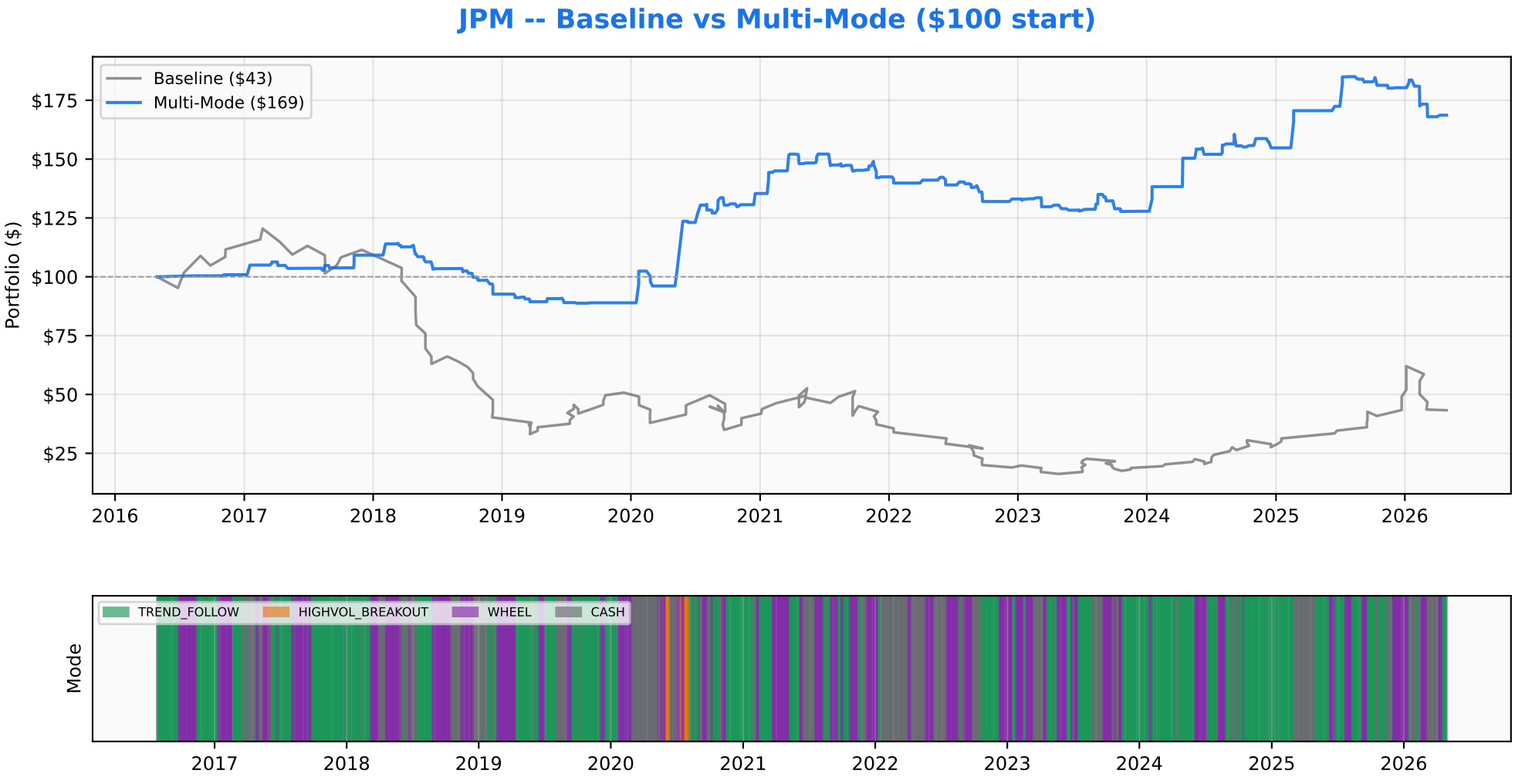
2022
2023
2024
2025
2026


SPY -- Baseline vs Multi-Mode ($100 start)
Baseline ($53)
Multi-Mode ($114)
Portfolio ($)
$120
$110
$100
$90
$80
$70
$60
$50
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH


QQQ -- Baseline vs Multi-Mode ($100 start)
Baseline ($176)
Multi-Mode ($143)
Portfolio ($)
$350
$300
$250
$200
$150
$100
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
Mode
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026


AMZN -- Baseline vs Multi-Mode ($100 start)
Baseline ($576)
Multi-Mode ($245)
Portfolio ($)
$700
$600
$500
$400
$300
$200
$100
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH


TSLA -- Baseline vs Multi-Mode ($100 start)
Baseline ($158)
Multi-Mode ($20028)
Portfolio ($)
$20000
$15000
$10000
$5000
$0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH


JPM -- Baseline vs Multi-Mode ($100 start)
Baseline ($43)
Multi-Mode ($169)
Portfolio ($)
$175
$150
$125
$100
$75
$50
$25
2016
2017
2018
2019
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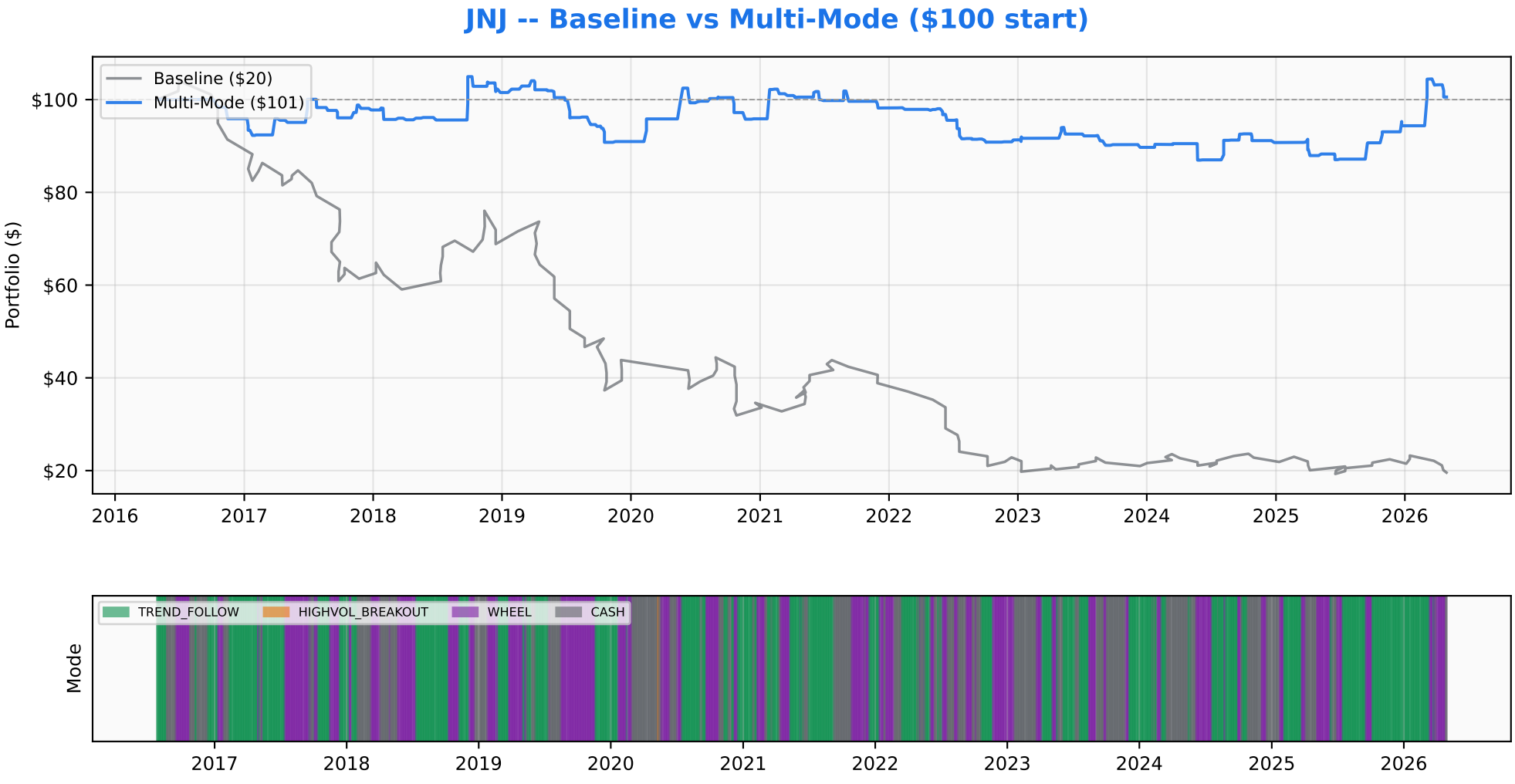
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH


JNJ -- Baseline vs Multi-Mode ($100 start)
Baseline ($20)
Multi-Mode ($101)
Portfolio ($)
$100
$80
$60
$40
$20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
2017
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2022
2023
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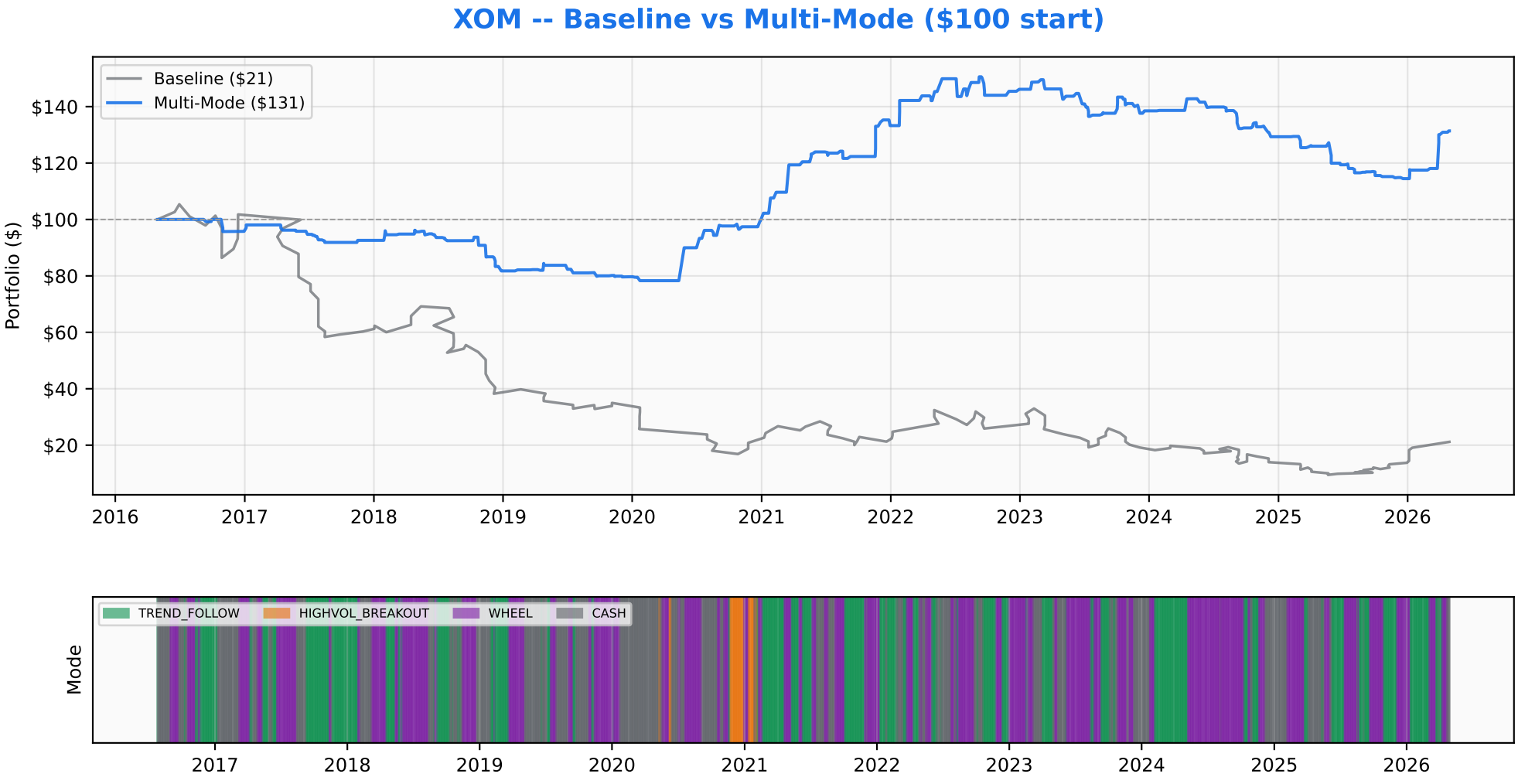
2025
2026


XOM -- Baseline vs Multi-Mode ($100 start)
Baseline ($21)
Multi-Mode ($131)
Portfolio ($)
$140
$120
$100
$80
$60
$40
$20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Mode
TREND_FOLLOW
HIGHVOL_BREAKOUT
WHEEL
CASH
2017
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2023
2024
2025
2026

Wheel Strategy Detail

Wheel Strategy Income Detail

Symbol	CSP	CC	Asgn	Called	Premium$
GOOGL	67	2	3	0	$ 5197
AAPL	61	1	1	0	$ 6004
MSFT	74	2	3	2	$ 11683
NVDA	63	8	10	2	$ 8375
SPY	44	0	1	0	$ 4512
QQQ	66	0	2	0	$ 7705
AMZN	75	4	5	1	$ 10595
TSLA	74	18	19	6	$ 60546
JPM	68	0	1	0	$ 4376
JNJ	57	1	1	0	$ 1479
XOM	72	2	2	0	$ 3109

Conclusion

Results Summary

```
Multi-Mode wins: 7/11 symbols
Baseline wins:   4/11 symbols
Avg Final Value: Old 456vsNew2295
Avg Sharpe:      Old 0.84 vs New 0.84
```

Per-Symbol Comparison

```
GOOGL  Old:772New:    268  (-504)  [TREND_FOLLOW]
AAPL   Old:188New:    344  (+156)  [TREND_FOLLOW]
MSFT   Old:2247New:    320  (-1927)  [TREND_FOLLOW]
NVDA   Old:760New:   3380  (+2620)  [TREND_FOLLOW]
SPY    Old:53New:    114  (+61)  [TREND_FOLLOW]
QQQ    Old:176New:    143  (-33)  [TREND_FOLLOW]
AMZN   Old:576New:    245  (-331)  [WHEEL]
TSLA   Old:158New:  20028  (+19871)  [CASH]
JPM    Old:43New:    169  (+125)  [TREND_FOLLOW]
JNJ    Old:20New:    101  (+81)  [TREND_FOLLOW]
XOM    Old:21New:    131  (+110)  [WHEEL]
```

```
Total backtest time: 24s
```